

Assignment #1

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Objectives

In this assignment, we review the knowledge we learned from Chapter 2 (EDA) lectures and skills we obtained from the Lab #1 and Lab #2 to answer practical questions with a real dataset.

I) Dataset Introduction

This is a census dataset about the adult income in the USA in the 1990s. The sample in this dataset was extracted by Barry Becker from the 1994 Census database and now is open source by the University of California, Irvine.

Here I provide a clean version. More details about the full dataset can be found at the [UCI repository](#)

Link download: [dataset](#)

age	workclass	fnlwgt	education	education.num	marital.status	occupation	relationship	race	sex	capital.gain	capital.loss	hours.per.week	native.country
82	Private	132870	HS-grad	9	Widowed	Exec-managerial	Not-in-family	White	Female	0	4356	18	United-States
54	Private	140359	7th-8th	4	Divorced	Machine-op-inspct	Unmarried	White	Female	0	3900	40	United-States
41	Private	264663	Some-college	10	Separated	Prof-specialty	Own-child	White	Female	0	3900	40	United-States
34	Private	216864	HS-grad	9	Divorced	Other-service	Unmarried	White	Female	0	3770	45	United-States
38	Private	150601	10th	6	Separated	Adm-clerical	Unmarried	White	Male	0	3770	40	United-States
74	State-gov	88638	Doctorate	16	Never-married	Prof-specialty	Other-relative	White	Female	0	3683	20	United-States
68	Federal-gov	422013	HS-grad	9	Divorced	Prof-specialty	Not-in-family	White	Female	0	3683	40	United-States
45	Private	172274	Doctorate	16	Divorced	Prof-specialty	Unmarried	Black	Female	0	3004	35	United-States
38	Self-emp-not-inc	164526	Prof-school	15	Never-married	Prof-specialty	Not-in-family	White	Male	0	2824	45	United-States
52	Private	129177	Bachelors	13	Widowed	Other-service	Not-in-family	White	Female	0	2824	20	United-States
32	Private	136204	Masters	14	Separated	Exec-managerial	Not-in-family	White	Male	0	2824	55	United-States
46	Private	45363	Prof-school	15	Divorced	Prof-specialty	Not-in-family	White	Male	0	2824	40	United-States
45	Private	172822	11th	7	Divorced	Transport-moving	Not-in-family	White	Male	0	2824	76	United-States
57	Private	317847	Masters	14	Divorced	Exec-managerial	Not-in-family	White	Male	0	2824	50	United-States
34	Private	203034	Bachelors	13	Separated	Sales	Not-in-family	White	Male	0	2824	50	United-States

Each entry contains the following information about an individual:

- **age**: the age of an individual (Integer > 0)
- **workclass**: a general term to represent the employment status of an individual (Private, Selfempnotinc, Selfempinc, Federalgov, Localgov, Stategov, Withoutpay, Neverworked)

- **fnlwgt**: final weight. In other words, this is the number of people the census believes the entry represents (Integer > 0)
- **education**: the highest level of education achieved by an individual (Bachelors, Somecollege, 11th, HSgrad, Profschool, Assocacdm, Assocvoc, 9th, 7th8th, 12th, Masters, 1st4th, 10th, Doctorate, 5th6th, Preschool).
- **education.num**: the highest level of education achieved in numerical form (Integer > 0)
- **marital.status**: marital status of an individual. Marriedcivspouse corresponds to a civilian spouse while MarriedAFspouse is a spouse in the Armed Forces (Marriedcivspouse, Divorced, Nevermarried, Separated, Widowed, Marriedspouseabsent, MarriedAFspouse).
- **occupation**: the general type of occupation of an individual (Techsupport, Craftrepair, Otherservice, Sales, Execmanagerial, Profspecialty, Handlerscleaners, Machineopinspct, Admclerical, Farmingfishing, Transportmoving, Privhouseserv, Protectiveserv, ArmedForces).
- **relationship**: represents what this individual is relative to others. For example an individual could be a Husband. Each entry only has one relationship attribute and is somewhat redundant with marital status (Wife, Ownchild, Husband, Notinfamily, Otherrelative, Unmarried)
- **race**: Descriptions of an individual's race (White, AsianPacIslander, AmerIndianEskimo, Other, Black)
- **sex**: the biological sex of the individual (Male, Female)
- **capital.gain**: capital gains for an individual (Integer > 0)
- **capital.loss**: capital loss for an individual (Integer > 0)
- **hours.per.week**: the hours an individual has reported to work per week (Integer > 0)
- **native.country**: country of origin for an individual (UnitedStates, Cambodia, England, PuertoRico, Canada, Germany, OutlyingUS(GuamUSVletc), India, Japan, Greece, South, China, Cuba, Iran, Honduras, Philippines, Italy, Poland, Jamaica, Vietnam, Mexico, Portugal, Ireland, France, DominicanRepublic, Laos, Ecuador, Taiwan, Haiti, Columbia, Hungary, Guatemala, Nicaragua, Scotland, Thailand, Yugoslavia, ElSalvador, Trinidad&Tobago, Peru, Hong, HolandNetherlands).

II. Questions

Questions about variables in general

Q1) What are the categorical variables in this dataset, and why? (10)

Q2) What are the quantitative variables in this dataset, and why? (10)

Questions about examining distribution of one categorical variable

Q3) Examine distribution of **workclass** variable (10)

- Create a frequency distribution table for different values of **workclass** variable
- Plot pie chart for **workclass** variable
- Plot bar chart for **workclass** variable
- Provide your summary/interpretation about the distribution of **workclass** variable.

Q4) Examine distribution of **education** variable (10)

- Create a frequency distribution table for different values of **education** variable
- Plot pie chart for **education** variable
- Plot bar chart for **education** variable
- Provide your summary/interpretation about the distribution of **education** variable.

Q5) Examine distribution of **marital.status** variable (10)

- Create a frequency distribution table for different values of **marital.status** variable
- Plot pie chart for **marital.status** variable
- Plot bar chart for **marital.status** variable
- Provide your summary/interpretation about the distribution of **marital.status** variable.

Q6) Examine distribution of **occupation** variable (10)

- Create a frequency distribution table for different values of **occupation** variable
- Plot pie chart for **occupation** variable
- Plot bar chart for **occupation** variable
- Provide your summary/interpretation about the distribution of **occupation** variable.

Q7) Examine distribution of **sex** variable (10)

- Create a frequency distribution table for different values of **sex** variable
- Plot pie chart for **sex** variable
- Plot bar chart for **sex** variable
- Provide your summary/interpretation about the distribution of **sex** variable.

Questions about examining distribution of one quantitative variable

Q8) Examine distribution of **age** variable (10)

- Plot histogram for **age** variable
- Interpret your observation from the histogram with respect to the shape, center, spread, and outliers.
- Calculate measures of center of the distribution of **age**: mode, mean, median

- Calculate measures of spread of the distribution of **age**: range, IQR, outliers based on 1.5IQR criterion, standard deviation.
- Plot the box plot for **age** variable
- Provide your summary/interpretation about the distribution of **age**.

Q9) Examine distribution of **education.num** variable (10)

- Plot histogram for **education.num** variable
- Interpret your observation from the histogram with respect to the shape, center, spread, and outliers.
- Calculate measures of center of the distribution of **education.num**: mode, mean, median
- Calculate measures of spread of the distribution of **education.num**: range, IQR, outliers based on 1.5IQR criterion, standard deviation.
- Plot the box plot for **education.num** variable
- Provide your summary/interpretation about the distribution of **education.num**.

Q10) Examine distribution of **hours.per.week** variable (10)

- Plot histogram for **hours.per.week** variable
- Interpret your observation from the histogram with respect to the shape, center, spread, and outliers.
- Calculate measures of center of the distribution of **hours.per.week**: mode, mean, median
- Calculate measures of spread of the distribution of **hours.per.week**: range, IQR, outliers based on 1.5IQR criterion, standard deviation.
- Plot the box plot for **hours.per.week** variable
- Provide your summary/interpretation about the distribution of **hours.per.week**.

III. Submissions

Submission is via Nexus. Due: 2025/01/20 @ 11:59 PM

One of the following formats is accepted for a submission:

- Jupyter notebook